Name of Student: Shreyasi Mishra	**PRN: 25070122294**
Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 4	
TITLE : Wrapper Classes in Java	

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Programme

1. **Create a program to convert student marks from String format to Integer and calculate total marks.**

```
public class StudentMarks {

   public static void main(String[] args) {

      String marks1 = "85";

      String marks2 = "90";

      String marks3 = "78";

      Integer m1 = Integer.valueOf(marks1);

      Integer m2 = Integer.valueOf(marks2);

      Integer m3 = Integer.valueOf(marks3);

      Integer total = m1 + m2 + m3;

      System.out.println("Student Marks");

      System.out.println("Subject 1: " + m1);

      System.out.println("Subject 2: " + m2);
```

```
        System.out.println("Subject 3: " + m3);
        System.out.println("Total Marks: " + total);
        int primitiveTotal = total.intValue();
        System.out.println("Primitive Total: " + primitiveTotal);
    }
}
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```
import java.util.Scanner;
public class EmployeePayroll {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Employee ID: ");
        String empIdStr = sc.nextLine();
        System.out.print("Enter Basic Salary: ");
        String basicSalaryStr = sc.nextLine();
        System.out.print("Enter Bonus Amount: ");
        String bonusStr = sc.nextLine();
        Integer empId = Integer.valueOf(empIdStr);
        Double basicSalary = Double.valueOf(basicSalaryStr);
        Double bonus = Double.valueOf(bonusStr);
        if (basicSalary < 0 || bonus < 0) {
            System.out.println("Invalid salary or bonus amount!");
        } else {
            Double netSalary = basicSalary + bonus;
            System.out.println("\nEmployee Details");
```

```
            System.out.println("Employee ID : " + empId);
            System.out.println("Basic Salary: " + basicSalary);
            System.out.println("Bonus Amount: " + bonus);
            System.out.println("Net Salary  : " + netSalary);
            double primitiveSalary = netSalary.doubleValue();
            System.out.println("Primitive Net Salary: " + primitiveSalary);
        }
        sc.close();
    }
}
```

Output

Output

```
Student Marks
Subject 1: 85
Subject 2: 90
Subject 3: 78
Total Marks: 253
Primitive Total: 253
```

Output

```
Enter Employee ID: 1234
Enter Basic Salary: 58292
Enter Bonus Amount: 2500

Employee Details
Employee ID : 1234
Basic Salary: 58292.0
Bonus Amount: 2500.0
Net Salary  : 60792.0
Primitive Net Salary: 60792.0
```

Conclusion

Through this assignment, we understood the concept of wrapper classes in Java and demonstrated how primitive data types such as int, float, double, and char can be converted into their corresponding wrapper objects like Integer, Float, Double, and Character through a process called boxing, and converted back to primitives through unboxing. We also explored the practical use of wrapper class methods such as parseInt(), valueOf(), and toString(), highlighting how wrapper classes enable primitives to be used in contexts that require objects, such as Java Collections and generics. Overall, this assignment reinforced the importance of wrapper classes as a bridge between primitive types and object-oriented functionality in Java, making programs more versatile and compatible with the broader Java ecosystem.